International Cybersecurity and Digital Forensics Academy

CCSA104 Security Policy Management

Course: Check Point Certified Security Administrator (CCSA)

Course Code: CCSA104

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Registration ID: C11/26/CCSA/17002

Completion Date: 6 November 2026

VM lunched: ONLY A-SMS (due to downloading link expiration)

PART A: PRACTICAL TASKS (70 MARKS)

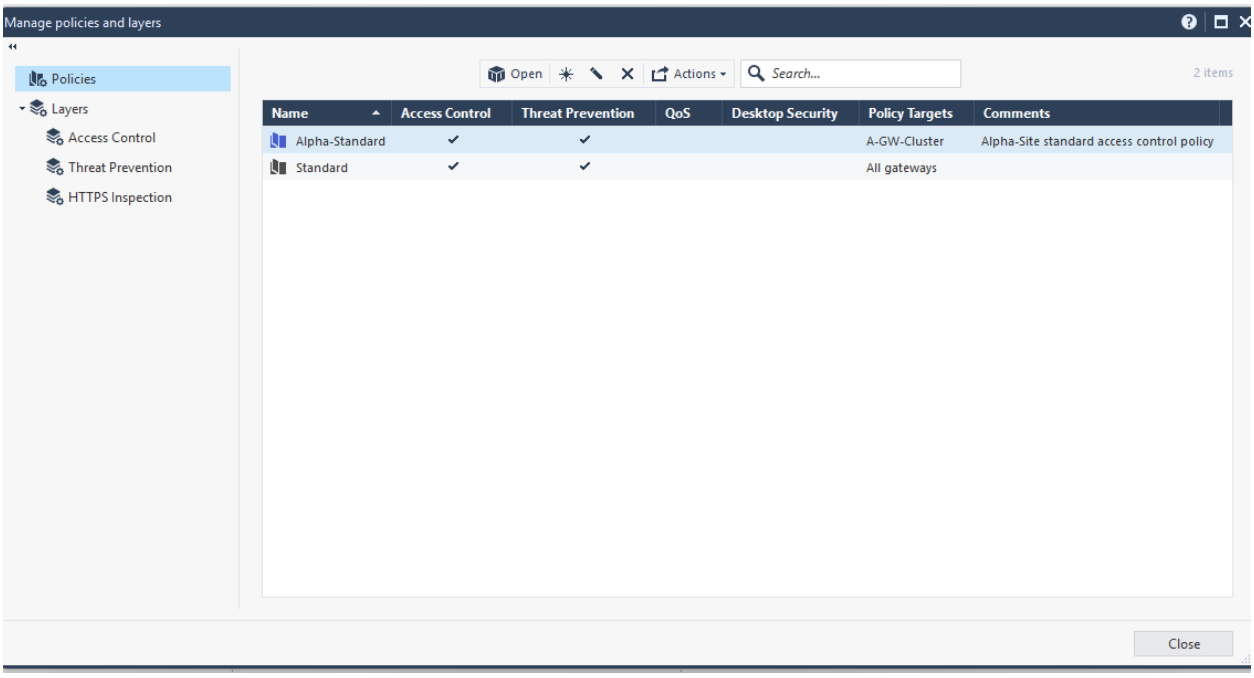
Task 1: Explore the Default Policy

The SmartConsole Access Control rulebase was opened and the default Cleanup Rule was inspected.

Policy Packages

The following policy packages were identified:

- Alpha-Standard
- Standard



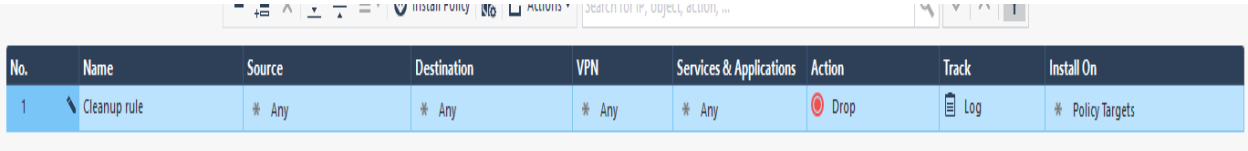
The active package name should be confirmed from the Security Policies view.

Default Cleanup Rule

Field	Observed value
Name	Cleanup Rule
Source	Any
Destination	Any
Services & Applications	Any

Action	Accept
Track	Log
Install On	Policy Targets

The original Cleanup Rule action was Accept. It was subsequently changed to Drop as required by the assignment.

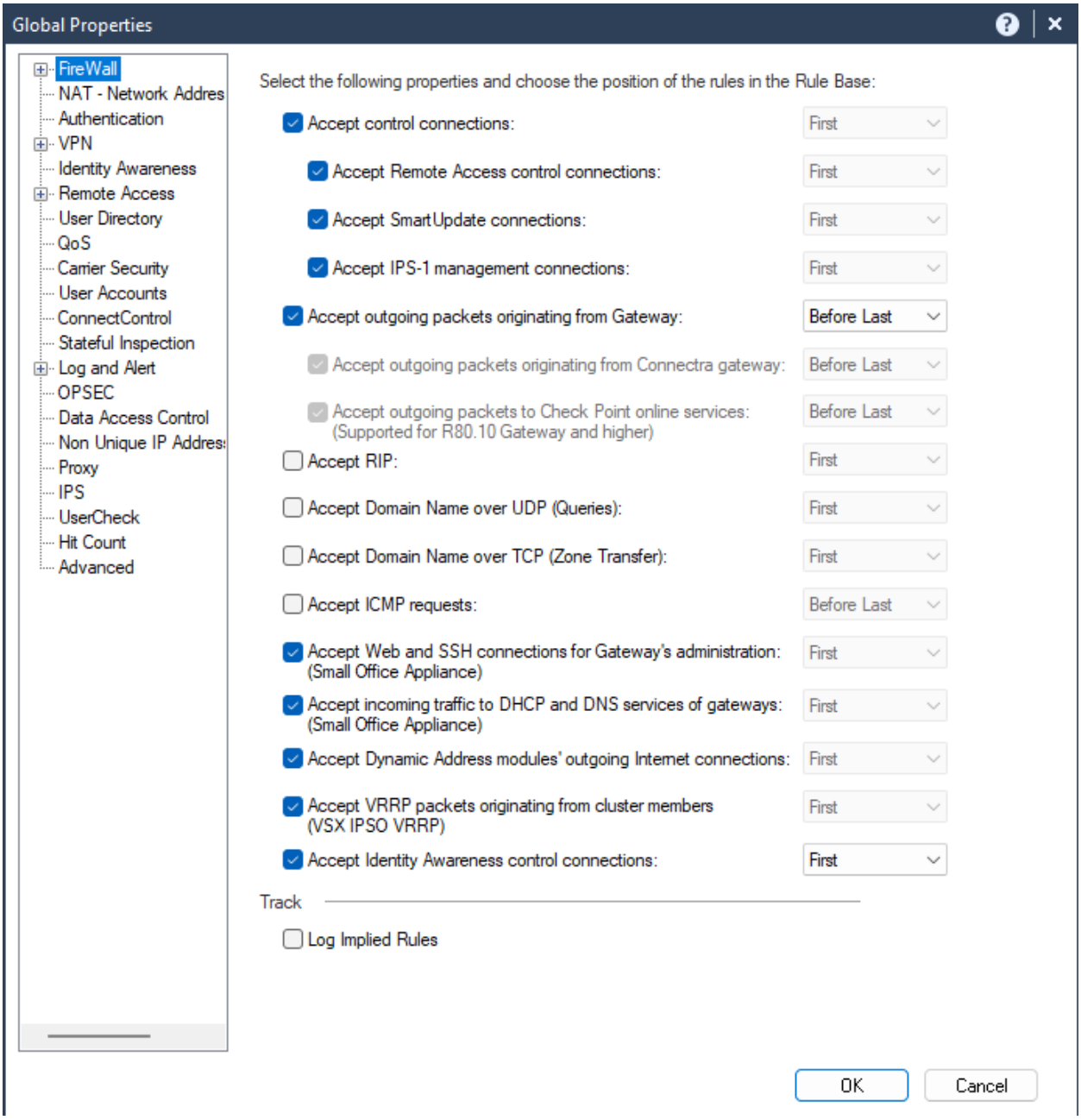


The Hits, Track and Install On columns were displayed. One existing column was hidden and displayed again.

Global Properties

Three categories identified were:

1. Firewall
2. VPN
3. Remote Access



Conclusion

This task provided experience in exploring the default security policy, inspecting Cleanup Rules, managing rulebase columns and identifying policy packages and Global Properties.

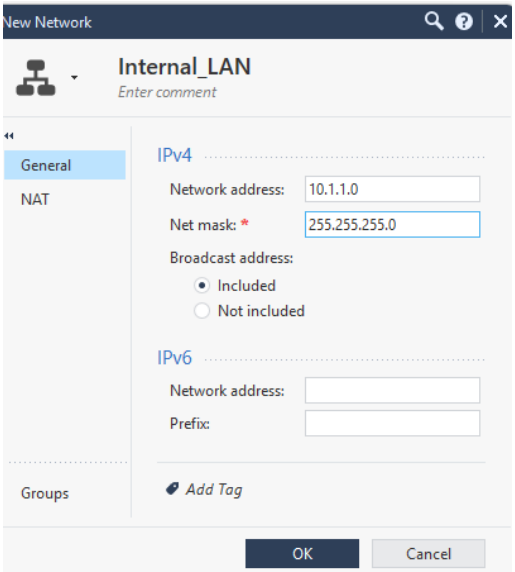
Task 2: Build the Access Control Rulebase

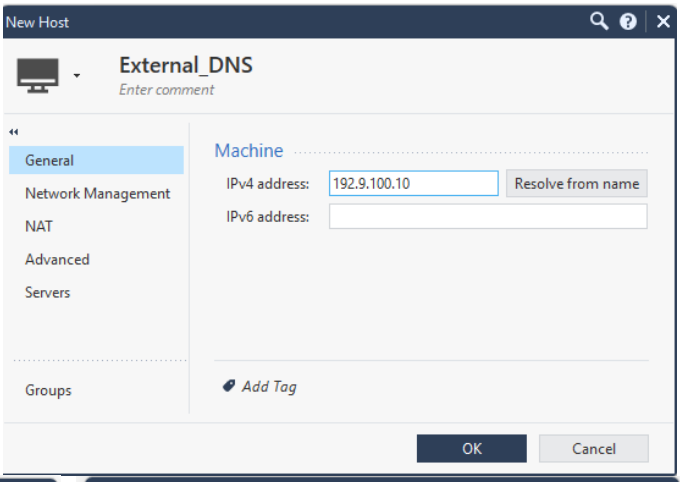
The required network and host objects were created in SmartConsole.

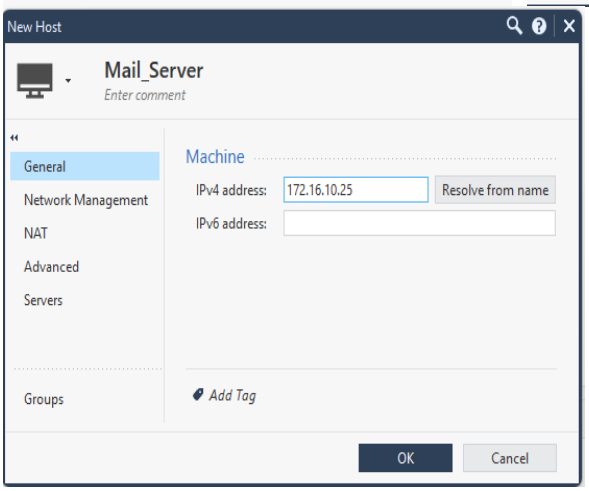
Network and Host Objects

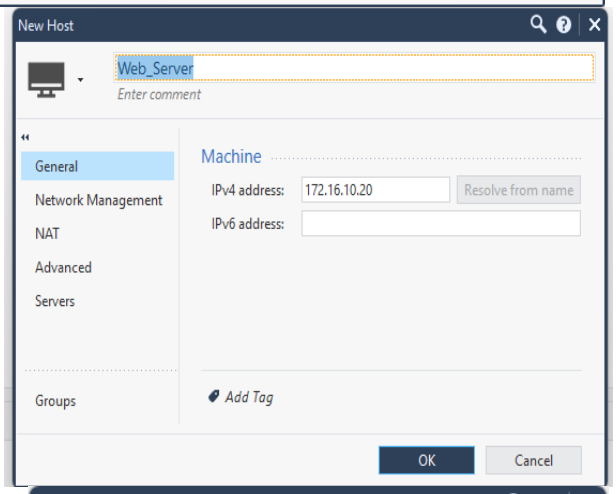
Object	Type	IP address
Internal_LAN	Network	10.1.1.0/24
DMZ_Net	Network	172.16.10.0/24
Admin_PC	Host	10.1.1.101
Web_Server	Host	172.16.10.20
Mail_Server	Host	172.16.10.25
External_DNS	Host	192.9.100.10

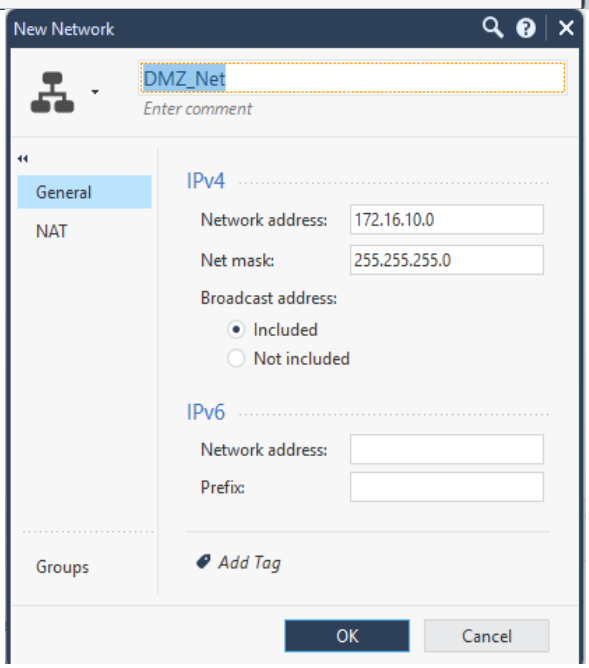
These are the example addresses used for the assignment.

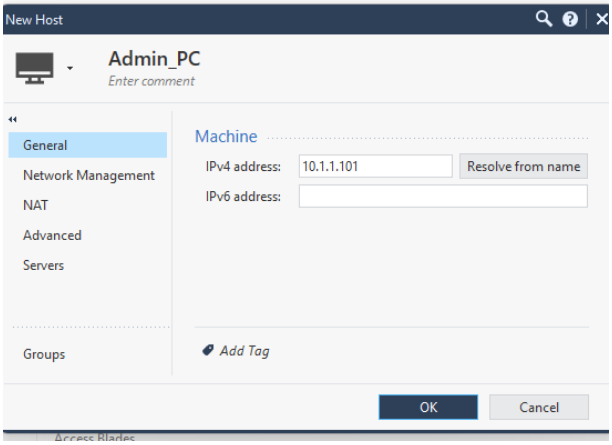

New Network
Internal_LAN
Enter comment
General
NAT
IPv4
Network address: 10.1.1.0
Net mask: * 255.255.255.0
Broadcast address:
Included
Not included
IPv6
Network address:
Prefix:
Groups
Add Tag
OK
Cancel


New Host
External_DNS
Enter comment
General
Network Management
NAT
Advanced
Servers
Machine
IPv4 address: 192.9.100.10
Resolve from name
IPv6 address:
Groups
Add Tag
OK
Cancel


New Host
Mail_Server
Enter comment
General
Network Management
NAT
Advanced
Servers
Machine
IPv4 address: 172.16.10.25
Resolve from name
IPv6 address:
Groups
Add Tag
OK
Cancel


New Host
Web_Server
Enter comment
General
Network Management
NAT
Advanced
Servers
Machine
IPv4 address: 172.16.10.20
Resolve from name
IPv6 address:
Groups
Add Tag
OK
Cancel


New Network
DMZ_Net
Enter comment
General
NAT
IPv4
Network address: 172.16.10.0
Net mask: 255.255.255.0
Broadcast address:
Included
Not included
IPv6
Network address:
Prefix:
Groups
Add Tag
OK
Cancel


New Host
Admin_PC
Enter comment
General
Network Management
NAT
Advanced
Servers
Machine
IPv4 address: 10.1.1.101
Resolve from name
IPv6 address:
Groups
Add Tag
OK
Cancel

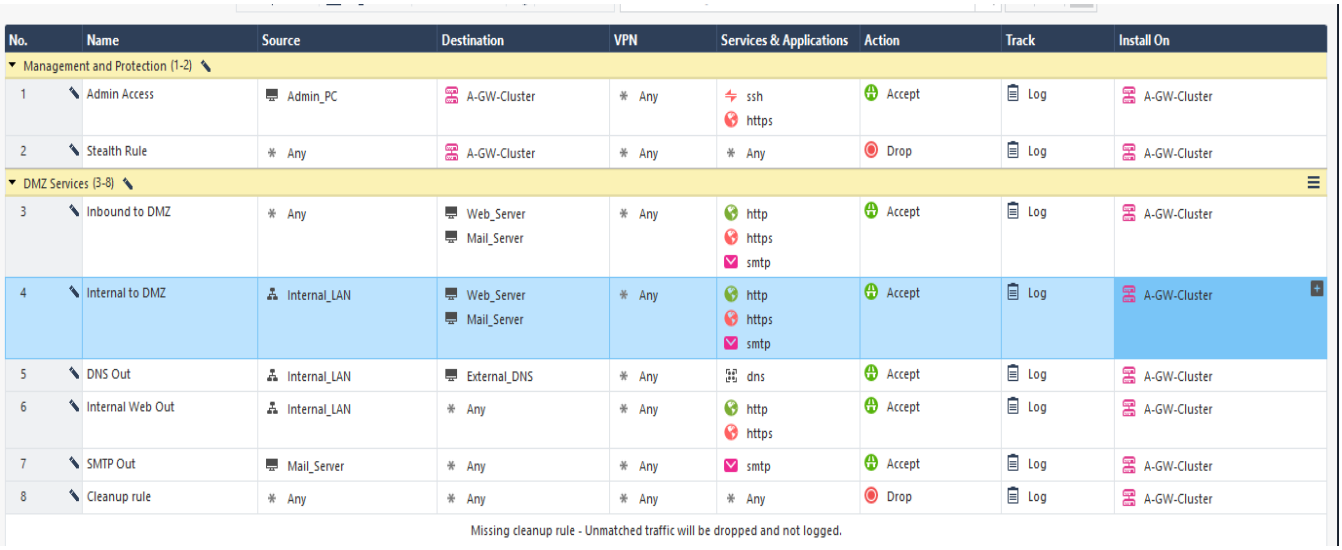
Security Gateway

Gateway object name: A-GW-Cluster.

Access Control Rules

The rules were configured as follows:

No.	Name	Source	Destination	Services	Action	Track
1	Admin Access	Admin_PC	A-GW-Cluster	ssh, https	Accept	Log
2	Stealth Rule	Any	A-GW-Cluster	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
4	Internal to DMZ	Internal_LAN	DMZ_Net	http, https, smtp	Accept	Log
5	DNS Out	Internal_LAN	External_DNS	dns	Accept	Log
6	Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log
8	Cleanup Rule	Any	Any	Any	Drop	Log



The DNS service was required to support both UDP and TCP.

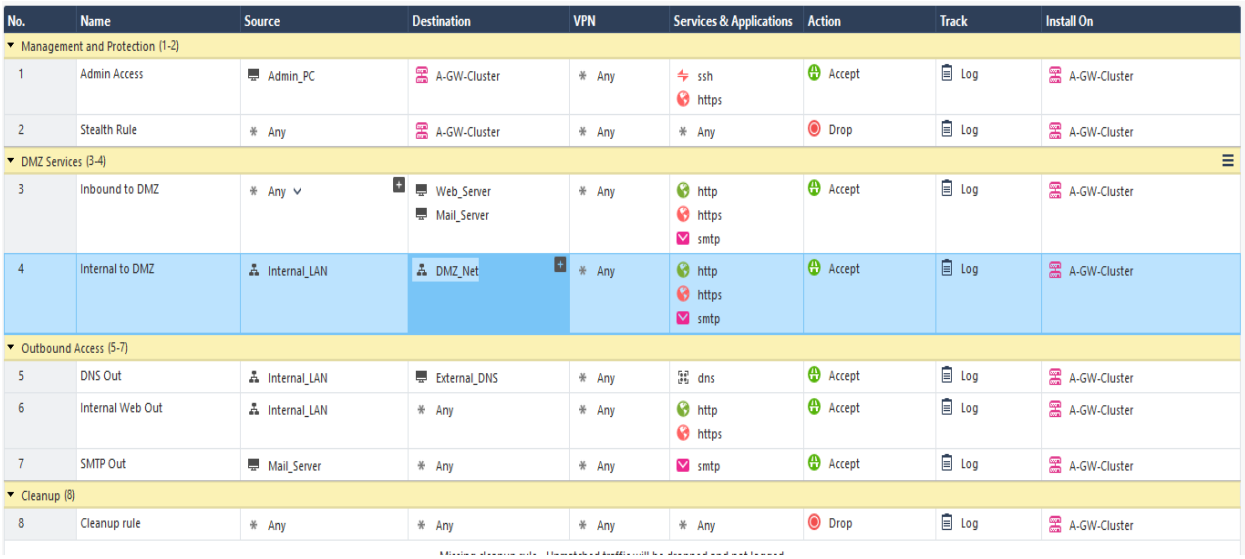
The original Cleanup Rule was modified instead of creating a new one.

Conclusion

The Access Control rulebase was created using the Positive Control Model. Only specifically permitted traffic was allowed, while other traffic was intended to be dropped and logged.

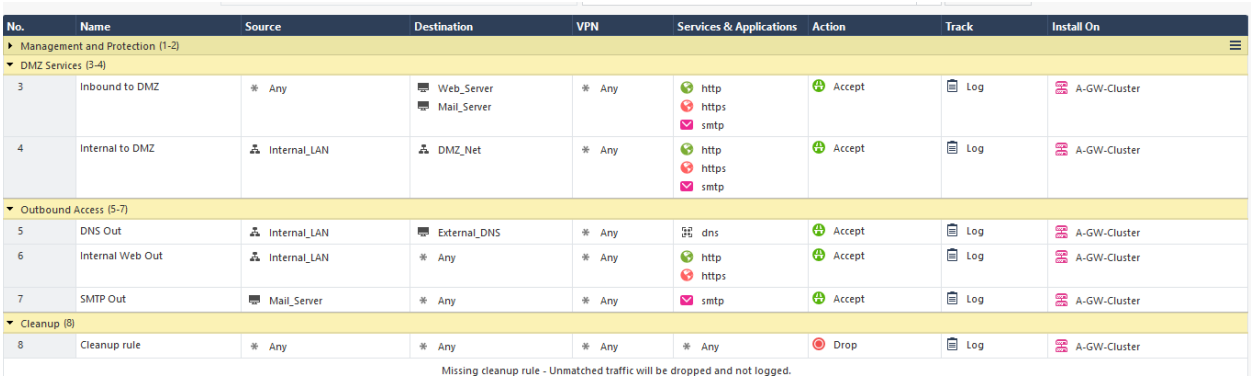
Task 4: Organize the Rulebase with Sections

Four sections were created in the Access Control rulebase.



No.	Name	Source	Destination	VPN	Services & Applications	Action	Track	Install On
Management and Protection (1-2)								
1	Admin Access	Admin_PC	A-GW-Cluster	Any	ssh https	Accept	Log	A-GW-Cluster
2	Stealth Rule	Any	A-GW-Cluster	Any	Any	Drop	Log	A-GW-Cluster
DMZ Services (3-4)								
3	Inbound to DMZ	Any	Web_Server Mail_Server	Any	http https smtp	Accept	Log	A-GW-Cluster
4	Internal to DMZ	Internal_LAN	DMZ_Net	Any	http https smtp	Accept	Log	A-GW-Cluster
Outbound Access (5-7)								
5	DNS Out	Internal_LAN	External_DNS	Any	dns	Accept	Log	A-GW-Cluster
6	Internal Web Out	Internal_LAN	Any	Any	http https	Accept	Log	A-GW-Cluster
7	SMTP Out	Mail_Server	Any	Any	smtp	Accept	Log	A-GW-Cluster
Cleanup (8)								
8	Cleanup rule	Any	Any	Any	Any	Drop	Log	A-GW-Cluster

Section	Rules
Management and Protection	Admin Access, Stealth Rule
DMZ Services	Inbound to DMZ, Internal to DMZ
Outbound Access	DNS Out, Internal Web Out, SMTP Out
Cleanup	Cleanup Rule



No.	Name	Source	Destination	VPN	Services & Applications	Action	Track	Install On
Management and Protection (1-2)								
DMZ Services (3-4)								
3	Inbound to DMZ	Any	Web_Server Mail_Server	Any	http https smtp	Accept	Log	A-GW-Cluster
4	Internal to DMZ	Internal_LAN	DMZ_Net	Any	http https smtp	Accept	Log	A-GW-Cluster
Outbound Access (5-7)								
5	DNS Out	Internal_LAN	External_DNS	Any	dns	Accept	Log	A-GW-Cluster
6	Internal Web Out	Internal_LAN	Any	Any	http https	Accept	Log	A-GW-Cluster
7	SMTP Out	Mail_Server	Any	Any	smtp	Accept	Log	A-GW-Cluster
Cleanup (8)								
8	Cleanup rule	Any	Any	Any	Any	Drop	Log	A-GW-Cluster

Missing cleanup rule - Unmatched traffic will be dropped and not logged.

Figure 1, management section collapsed

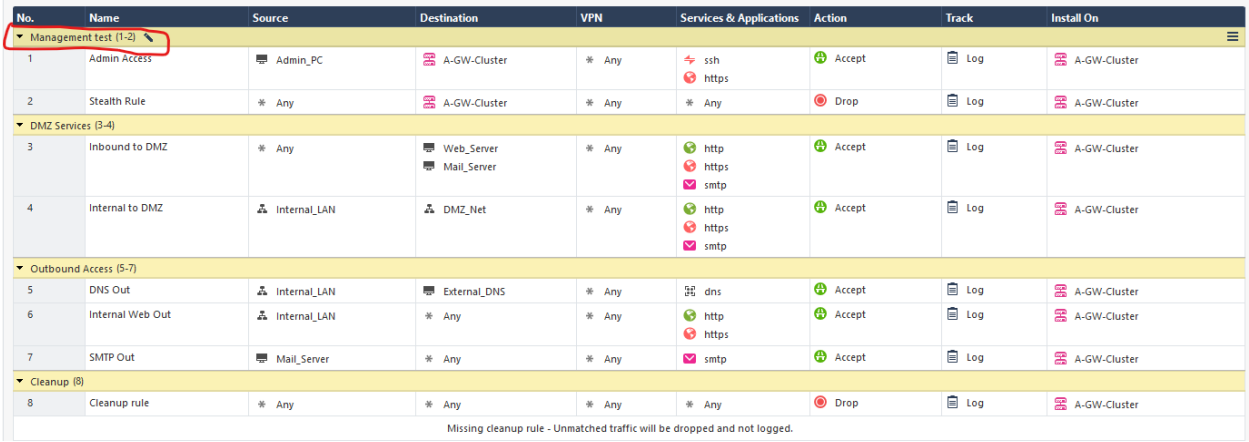


Figure 2, management section changed to test

The sections were collapsed and expanded. One section title was temporarily edited and then restored.

Explanation

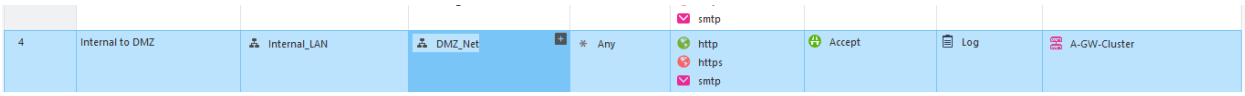
Sections are visual organisational divisions in SmartConsole. They do not affect rule enforcement order and are not sent to the Security Gateway as enforcement rules.

Task 5: Internal to DMZ and Security Zones

The Internal to DMZ rule was added with the following configuration:

Field	Value
Name	Internal to DMZ
Source	Internal_LAN
Destination	DMZ_Net
Services	http, https, smtp
Action	Accept
Track	Log

The rule was placed in the DMZ Services section.



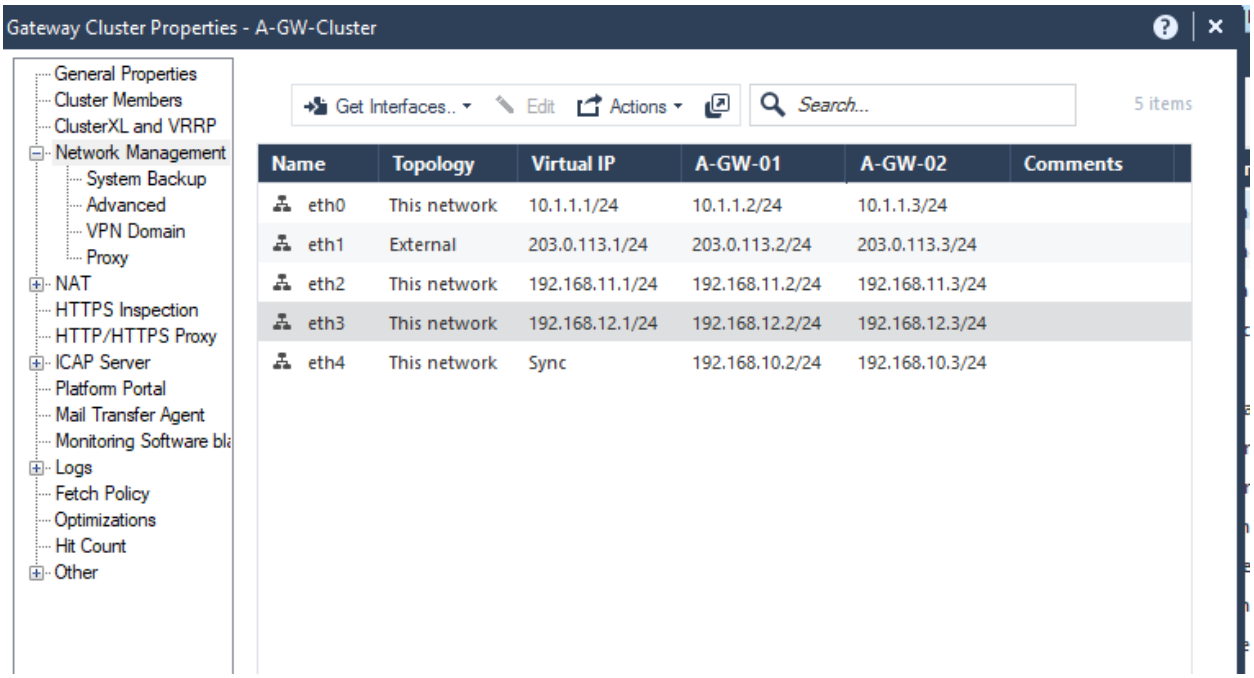
Gateway Interface Inspection

The Gateway object A-GW-Cluster was inspected.

The cluster had two members:

- A-GW-01
- A-GW-02

Five interfaces, eth0 through eth4, were identified.



The interface topology and IP addresses were inspected. No Security Zone configuration changes were made.

Conclusion

The Internal to DMZ rule was created and published. Gateway interfaces were inspected to understand the network topology.

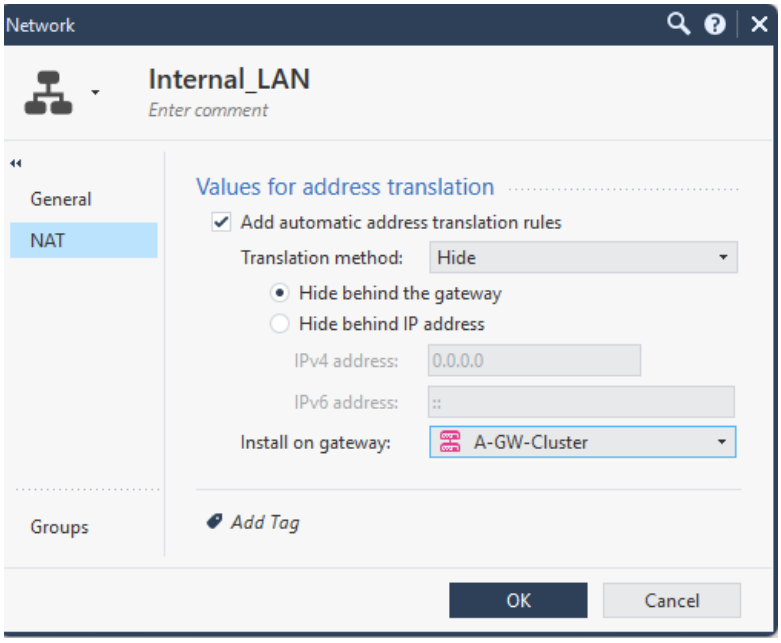
Task 6: Configure Hide NAT

The Internal_LAN network object was opened and automatic address translation was enabled.

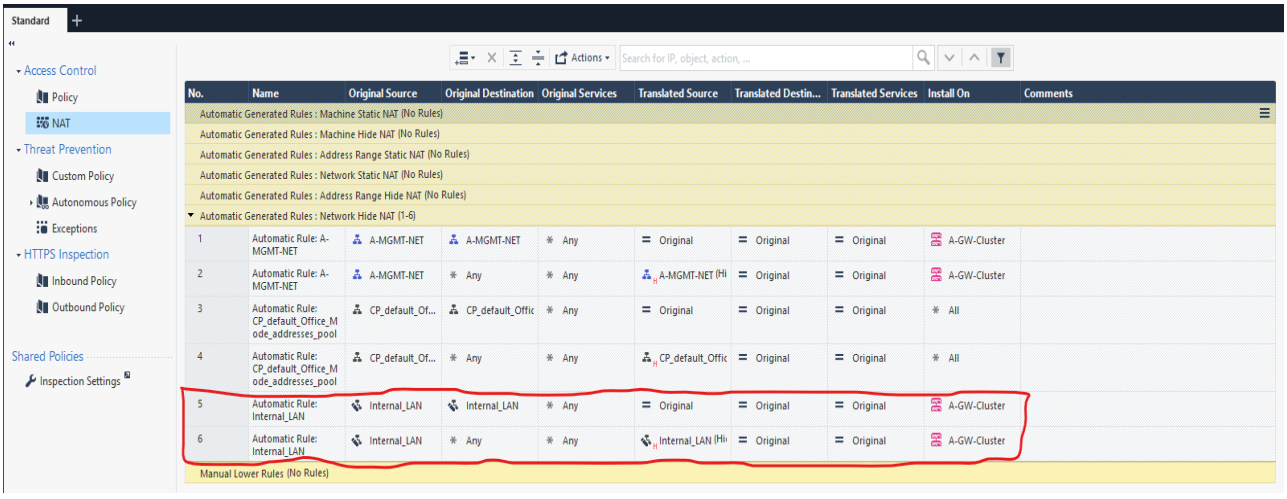
NAT Configuration

Setting	Configuration
Object	Internal_LAN
Automatic NAT	Enabled
Translation Method	Hide
Hide Behind	Gateway
Gateway	A-GW-Cluster

The changes were published successfully.



The NAT rulebase was opened and the automatically generated NAT rule for Internal_LAN was located.



Explanation of Automatic NAT

This is an automatic NAT rule because it is generated by enabling address translation directly in the network object's NAT settings. SmartConsole automatically creates the corresponding NAT rule instead of requiring the administrator to create it manually.

Explanation of Many-to-One Translation

Many-to-one translation allows multiple internal computers to share one translated IP address when communicating with external networks. This reduces the number of public IP addresses required.

Source Port Translation

The Security Gateway uses Port Address Translation (PAT). It assigns or adjusts source port numbers when necessary to distinguish connections from different internal computers sharing the same translated IP address.

Task 7: Rule Order Experiment and Installation History

A temporary rule named TEMP Block Internal was created.

Temporary Rule Configuration

Field	Value
Name	TEMP Block Internal
Source	Internal_LAN
Destination	Any
Services	Any
Action	Drop
Track	Log

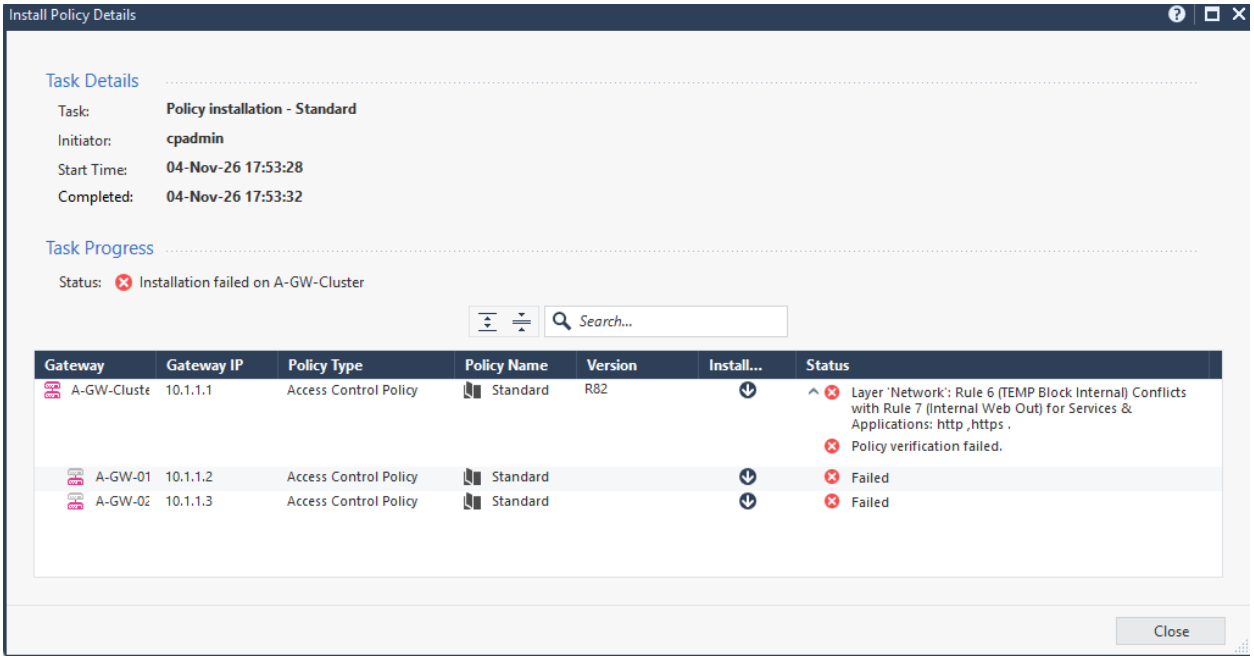
First Experiment

The temporary blocking rule was placed above Internal Web Out.



The rulebase demonstrated that a broad blocking rule can match before a more specific allowing rule.

This installation attempt resulted failure due to policy conflict General rule above a specific rule



Second Experiment

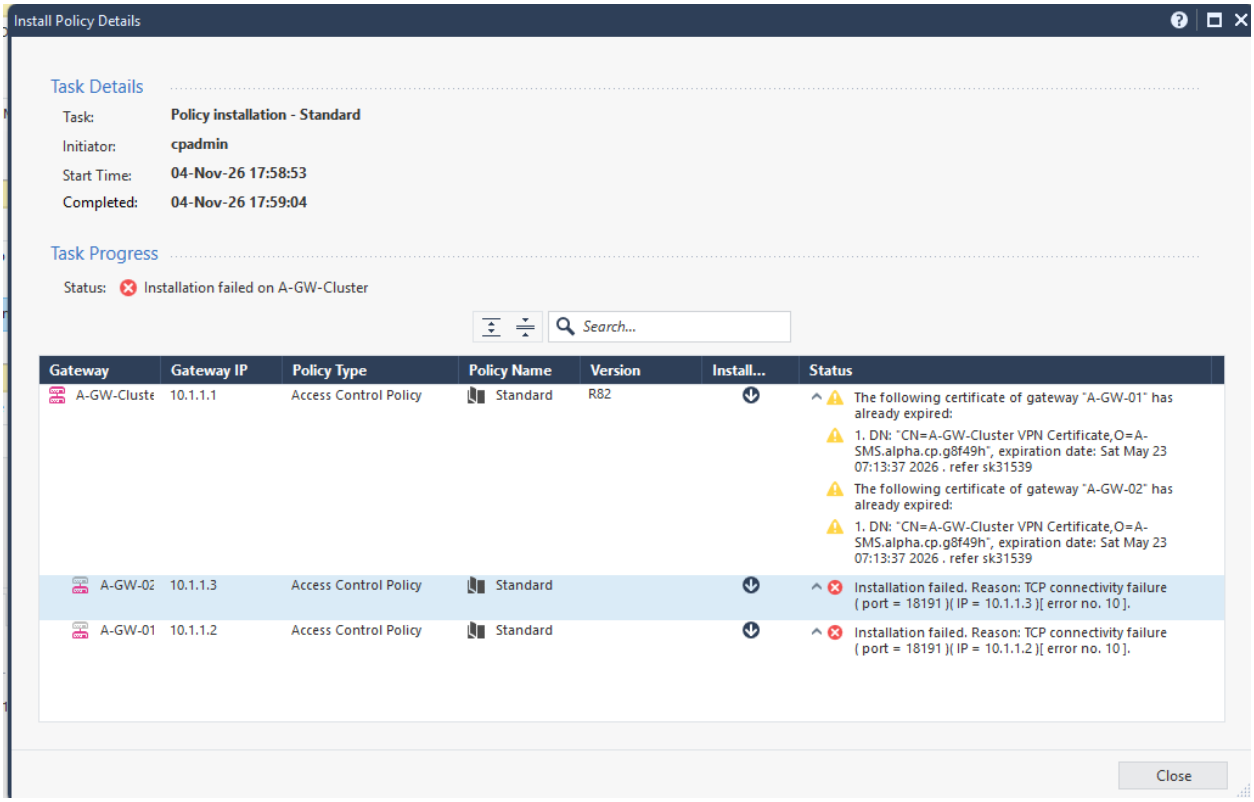
The temporary rule was moved below Internal Web Out.



This demonstrated that the more specific web-access rule takes precedence when it appears first.

Installation Attempt

Policy installation was attempted after publishing the changes on the second experiment.



However, installation was unsuccessful because:

- The Gateway certificate had expired (warning).
- TCP connectivity to the Gateway failed.

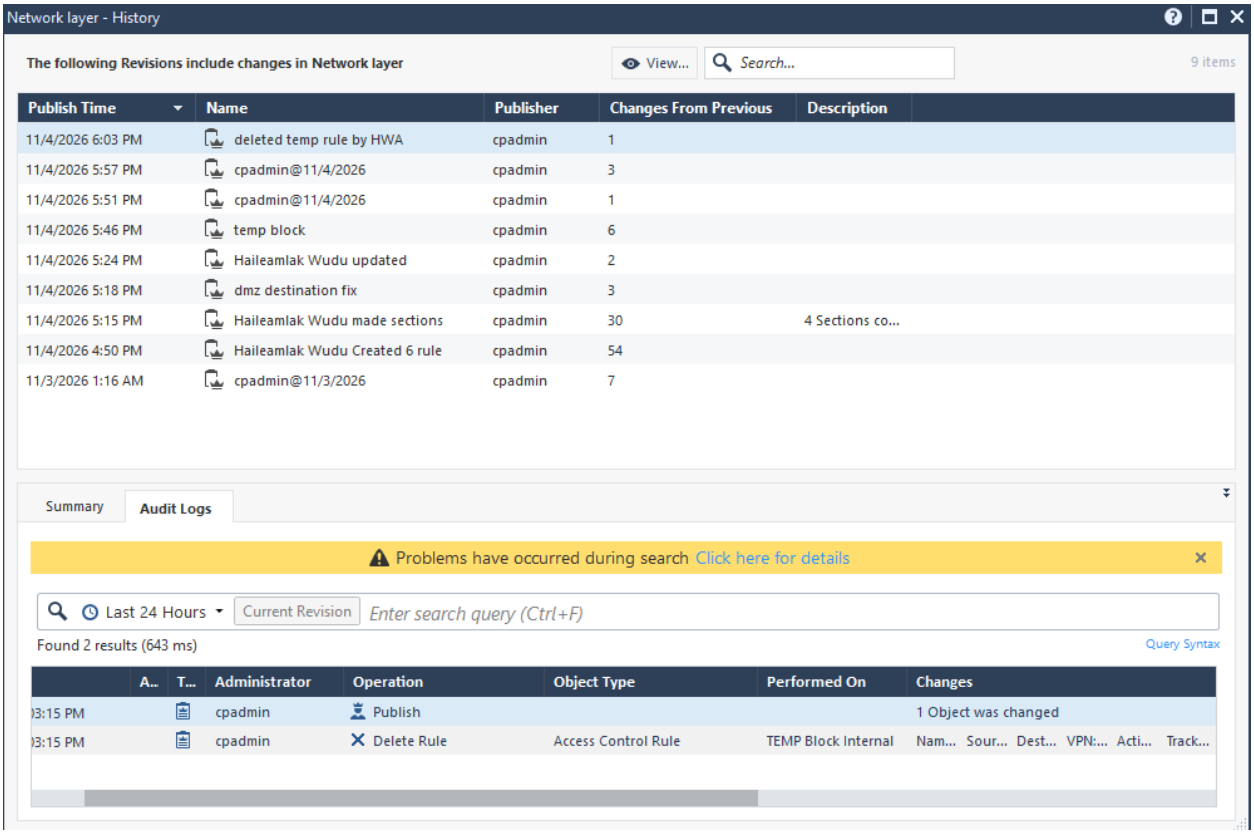
Therefore, successful policy installation and enforcement could not be verified due to **No Gateway virtual machines were used** not because configuration issue.

Removing the Temporary Rule

TEMP Block Internal was deleted after the experiment. The final intended rulebase was restored and the changes were published.

Installation History and Audit

The audit history was inspected to review the configuration changes. The installation attempt must be recorded as unsuccessful rather than successful.



Explanation of Rule Order

Firewall rules are evaluated from top to bottom. The first matching rule determines the action taken by the Security Gateway. Specific rules should be placed above general rules because a general blocking rule placed first can prevent legitimate traffic from reaching the intended allowing rules.

PART B: THEORY QUESTIONS (30 MARKS)

Section B1: Multiple Choice

Question	Correct answer
Q1	B
Q2	A
Q3	C
Q4	D
Q5	B

Section B2: Short Answers

Q6. Distinguish between Explicit Rules and Implied Rules.

Explicit rules are rules manually created by an administrator in the Access Control rulebase. They define which traffic is allowed or blocked.

Implied rules are automatically generated by Check Point based on predefined security settings. They are configured through Global Properties and their placement can be controlled through the implied-rule settings.

The Implicit Cleanup Action is the default action applied to traffic that does not match an applicable rule. Normally, unmatched traffic is dropped.

Q7. Explain why rule order matters in a rulebase.

Rule order matters because the Security Gateway evaluates rules from top to bottom and applies the first matching rule.

For example, TEMP Block Internal was placed above Internal Web Out. The blocking rule matched first and prevented internal web traffic from reaching the allowing rule. Moving the temporary rule below Internal Web Out demonstrated the importance of correct rule ordering.

Q8. Explain the difference between saving and publishing changes in SmartConsole.

Saving preserves changes in the administrator's management session, while publishing makes the changes available as an updated management configuration.

Unpublished changes are not included in a policy installation.

Many organisations publish during approved change windows to control configuration changes, coordinate administrators and reduce operational risks.

Q9. Compare Global Properties with Policy Packages.

Global Properties contain general security management settings that can affect policy behaviour, such as firewall and VPN-related settings. They can be opened from the Applications menu in SmartConsole.

Policy Packages contain the actual security rules and policy configuration applied to selected gateways. They can be accessed through Security Policies and Manage Policies and Layers.

Q10. Name the four predefined Security Zones.

The four predefined Security Zones are:

1. Internal

2. External
3. DMZ
4. VPN

Two rules for assigning interfaces are:

- An interface should be assigned to the zone representing the network connected to it.
- Interface topology and connected network information must be configured correctly to avoid incorrect security classification.

One benefit of zone-based rules is that administrators can write policies based on security areas instead of specifying every individual network address.

Section B3: Scenario Questions

Q11. An administrator inserts an Any–Any–Any Accept rule above the Stealth Rule.

An Any–Any–Any Accept rule above the Stealth Rule creates two security problems.

1. It allows traffic from any source to any destination using any service, which can expose the network to unauthorized access.
2. It can bypass the intended security controls because the firewall processes rules from top to bottom and applies the first matching rule.

The administrator should remove the unnecessary broad rule and replace it with specific rules that permit only required traffic.

The only type of rule normally allowed above the Stealth Rule is a specific Management rule, such as Admin Access, because administrators need controlled access to manage the Security Gateway.

Q12. Staff cannot access the Internet after Hide NAT is configured.

(a) Two benefits of NAT

1. NAT hides internal private IP addresses from external networks, providing an additional layer of network privacy.
2. Hide NAT allows multiple internal computers to share one public IP address, reducing the need for multiple public IP addresses.

(b) Three troubleshooting checks

1. **Check NAT configuration:** Verify that automatic Hide NAT is enabled for Internal_LAN and that it is hidden behind the correct Gateway. This ensures internal traffic receives the correct translated source address.

2. **Check the Access Control policy:** Verify that outbound traffic such as HTTP, HTTPS and DNS is allowed by the appropriate rules. If the firewall blocks the traffic, users cannot access the Internet even when NAT is configured correctly.
3. **Check Gateway connectivity and routing:** Verify that the Gateway is running, its external interface is configured correctly, and it has a valid default route to the Internet. This ensures translated traffic can reach external networks.